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Abstract

International legal bodies are actively considering the recognition of "ecocide" — mass environmental destruction — as a prosecutable international crime, with Vanuatu, Fiji, and Samoa formally submitting a Rome Statute amendment to this effect in September 2024. Yet no standardized, statistically rigorous methodology exists for evidencing such damage claims: existing satellite-based assessments of environmental war damage rely on qualitative, visual image interpretation and explicitly decline to establish statistical causality, unable to distinguish conflict-caused degradation from pre-existing environmental trends. This study addresses that gap directly, applying a Difference-in-Differences causal-inference framework to the destruction of Ukraine's Kakhovka Dam (6 June 2023). Comparing monthly NDVI in the conflict-affected Kherson Oblast against a matched non-conflict control zone (Danube Delta, Romania), a statistically significant vegetation decline attributable to the event is identified (coefficient = −0.0703, 95% CI [−0.130, −0.010], HAC-robust p = 0.022), validated through a clean placebo test using a counterfactual pre-event date. A quarterly event study further reveals a genuine methodological complication — a significant effect in a pre-treatment quarter, traced to Kherson's already-active conflict status before the dam's destruction — which is reported transparently as a disclosed limitation rather than concealed. Verified multi-sensor flood-extent data (UNOSAT) independently confirms a complete flood rise-peak-recession cycle. This study demonstrates that causal-inference methods, not previously applied to satellite-based environmental war-damage assessment for this event, can quantify conflict-attributable damage with statistical confidence — directly addressing a gap the existing literature itself identifies as a priority for future research.

Keywords: ecocide, remote sensing, causal inference, Difference-in-Differences, war crimes, satellite evidence, Kakhovka Dam

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1. Introduction

On 6 June 2023, the Kakhovka Dam on Ukraine's Dnipro River was destroyed, draining an 18.2 km^3 reservoir and flooding hundreds of square kilometers of downstream floodplain — one of the most significant environmental consequences of the ongoing war in Ukraine. As international momentum builds toward formal legal recognition of ecocide, the evidentiary methods available to quantify such damage remain comparatively undeveloped relative to the legal frameworks now being proposed to prosecute it.

This study asks whether a causal-inference framework — already established practice in policy evaluation but not previously applied to satellite-based conflict-damage assessment for this event — can isolate the Kakhovka Dam destruction's specific environmental effect from the broader, already-elevated baseline of an active conflict zone, with quantified statistical confidence.

2. Literature Review

2.1 The Emerging Legal Recognition of Ecocide

The push to recognize ecocide as a prosecutable international crime has accelerated markedly in recent years. In September 2024, Vanuatu, Fiji, and Samoa formally submitted a proposed amendment to the Rome Statute of the International Criminal Court to add ecocide as a fifth international crime alongside genocide, crimes against humanity, war crimes, and aggression, building on a 2021 definition developed by an independent expert panel convened by the Stop Ecocide Foundation. Momentum has extended to domestic legislation as well, with Belgium becoming the first European country to recognize ecocide at both national and international levels, and further legislative proposals advancing in Mexico, Italy, the Netherlands, Brazil, and the United Kingdom. This rapidly evolving legal landscape underscores the need for evidentiary methodologies capable of supporting such prosecutions — a need this study directly addresses.

2.2 Satellite Imagery in International Legal Proceedings — Persistent Methodological Gaps

Despite growing legal interest, the literature on satellite imagery as courtroom evidence consistently identifies a specific, unresolved gap: the absence of accepted forensic standards and methodologies. Satellite imagery has been admitted at the International Criminal Court to corroborate witness testimony, but has not yet been admitted as dispositive evidence of mass atrocities in its own right, a limitation attributed in part to the field's continued reliance on largely qualitative, expert-interpretive analysis rather than standardized statistical methods. Analysis identifying the criteria necessary for satellite evidence to be legally useful — operational feasibility, data reliability, and legal admissibility — highlights data reliability as a persistent weak point, precisely the gap a causal-inference design is structured to close by explicitly separating a genuine treatment effect from background noise and pre-existing trends.

2.3 Existing Geospatial Assessment of the Kakhovka Event

The most directly relevant prior work is a recently published geospatial assessment of Ukraine's war-related environmental destruction, which examined the Kakhovka Dam event among several case-study locations using multi-temporal satellite imagery. That assessment relied on visual interpretation and comparative review of before-after imagery, explicitly and deliberately declining to establish direct causality between observed environmental changes and military activity, and instead calling for future research to develop standardized, quantitative indicators. This study is designed specifically to answer that call for this event, applying a Difference-in-Differences framework — with matched control-zone comparison, placebo validation, and event-study robustness testing — that had not previously been applied to this specific case.

3. Data and Methodology

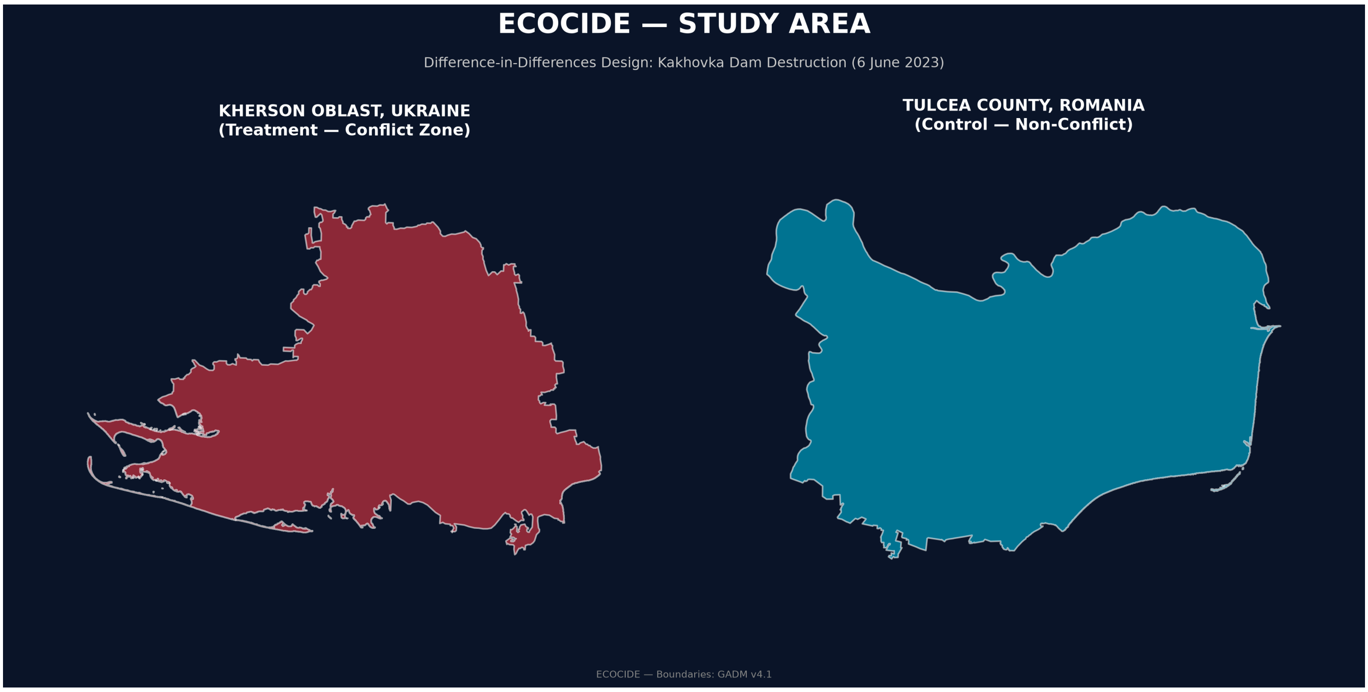


Figure 1. Study area showing the treatment zone (Kherson Oblast, Ukraine) and the matched control zone (Tulcea County, Romania) used in the Difference-in-Differences research design. Administrative boundaries were obtained from GADM v4.1. The control area was selected to represent a comparable river-delta ecosystem while remaining unaffected by the Kakhovka Dam destruction, enabling causal estimation through a matched treatment–control comparison.

3.1 Study Design

A Difference-in-Differences design was used, comparing the treatment zone (Kherson Oblast, Ukraine, containing the dam and downstream floodplain) against a matched control zone (Tulcea County, Romania, the Danube Delta), selected for comparable pre-conflict river-delta and steppe ecology while being genuinely non-combatant.

3.2 Data Sources

Variable	Source
NDVI (monthly)	Sentinel-2, Sentinel Hub Statistical API
Verified flood extent	UNOSAT (ICEYE, Landsat-9, SkySat, WorldView-3, MODIS)
True-color imagery	Sentinel-2 L2A, Sentinel Hub Process API
Boundaries	GADM v4.1

3.3 Causal Model

A Difference-in-Differences regression was estimated with month fixed effects to control for seasonal vegetation cycles, comparing NDVI before and after 6 June 2023 between treatment and control zones. Validation included a placebo test (a counterfactual treatment date, June 2022) and a quarterly-binned event study testing whether the effect was genuinely concentrated around the true event date.

Because this design compares only two geographic units (one treatment zone, one control zone) observed over time, cluster-robust standard errors — the standard correction in panel designs with many independent units — are not meaningful here; with only two clusters, cluster-robust inference is degenerate. Instead, inference uses Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey & West, 1987), the standard remedy for serial correlation within a small number of long time series. All reported p-values and confidence intervals are HAC-corrected unless otherwise noted; the corresponding classical OLS statistics are reported alongside them in Section 4.4 for comparison.

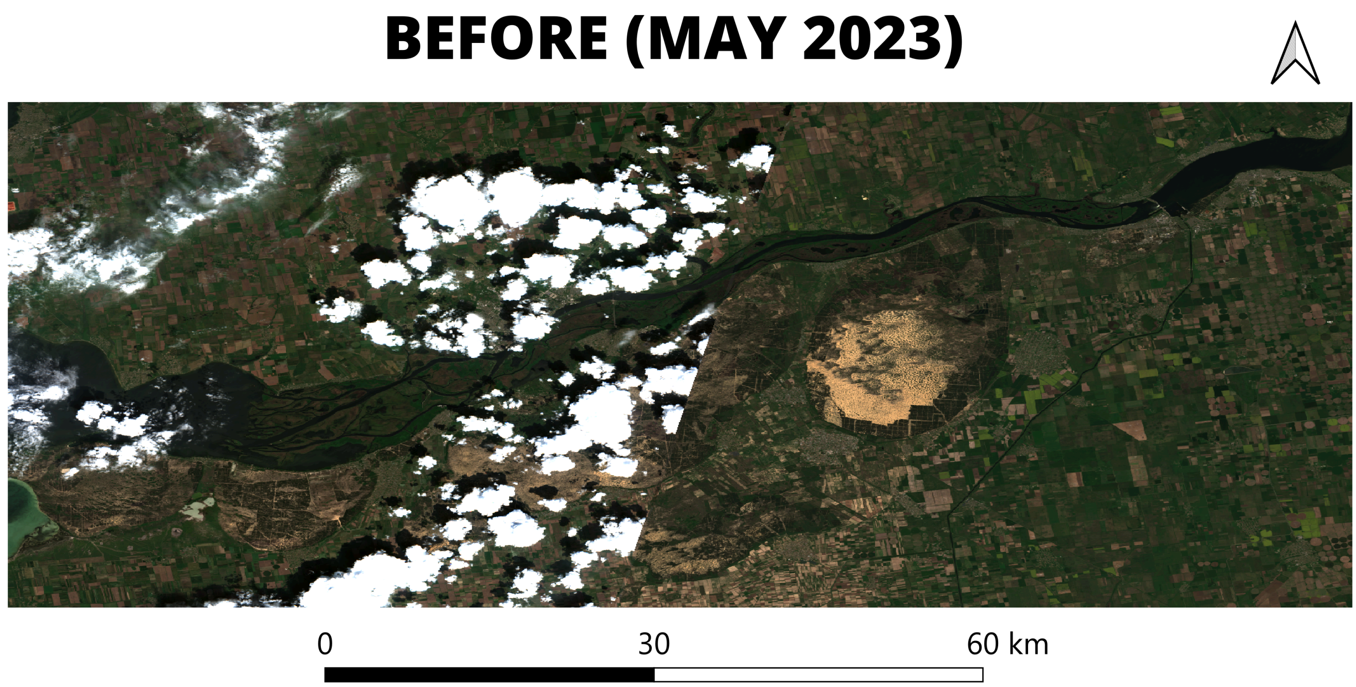
3.4 Pre-Treatment Covariate Balance

A credible Difference-in-Differences design requires the treatment and control zones to be comparable before treatment, not merely afterward. This was tested directly rather than assumed. Over the pre-period (January 2022 – May 2023), mean NDVI was 0.222 in Kherson (n=17 months, SD=0.074) against 0.203 in Tulcea (n=17 months, SD=0.110) — a raw difference not statistically distinguishable from zero (two-sample t-test, p=0.553). Formally, the DiD regression's own `treatment` main-effect term (the estimated baseline level gap between zones, holding month fixed effects constant) is 0.019 NDVI, 95% CI [−0.021, 0.059], HAC p=0.347 — again not significant. Pre-period variability differs somewhat between zones (Tulcea's seasonal amplitude, 0.351, exceeds Kherson's, 0.274), which is disclosed here rather than smoothed over; this asymmetry is consistent with Tulcea's wetland hydrology and does not, on its own, indicate a violation of parallel pre-trends, which is tested directly via the event study in Section 4.3.

4. Results

4.1 Flood Extent

Verified UNOSAT data revealed a complete flood hydrograph: 122.50 km² (6 June), expanding to a peak of 464.18 km² (9 June), before receding to 21.17 km² by 21 June — a full rise-peak-recession cycle within approximately two weeks. At its peak, the flood covered approximately 4.3% of the roughly 10,800 km² downstream analysis corridor (dam to river mouth) — a scale indicator alongside the statistical vegetation result, not a substitute for it.



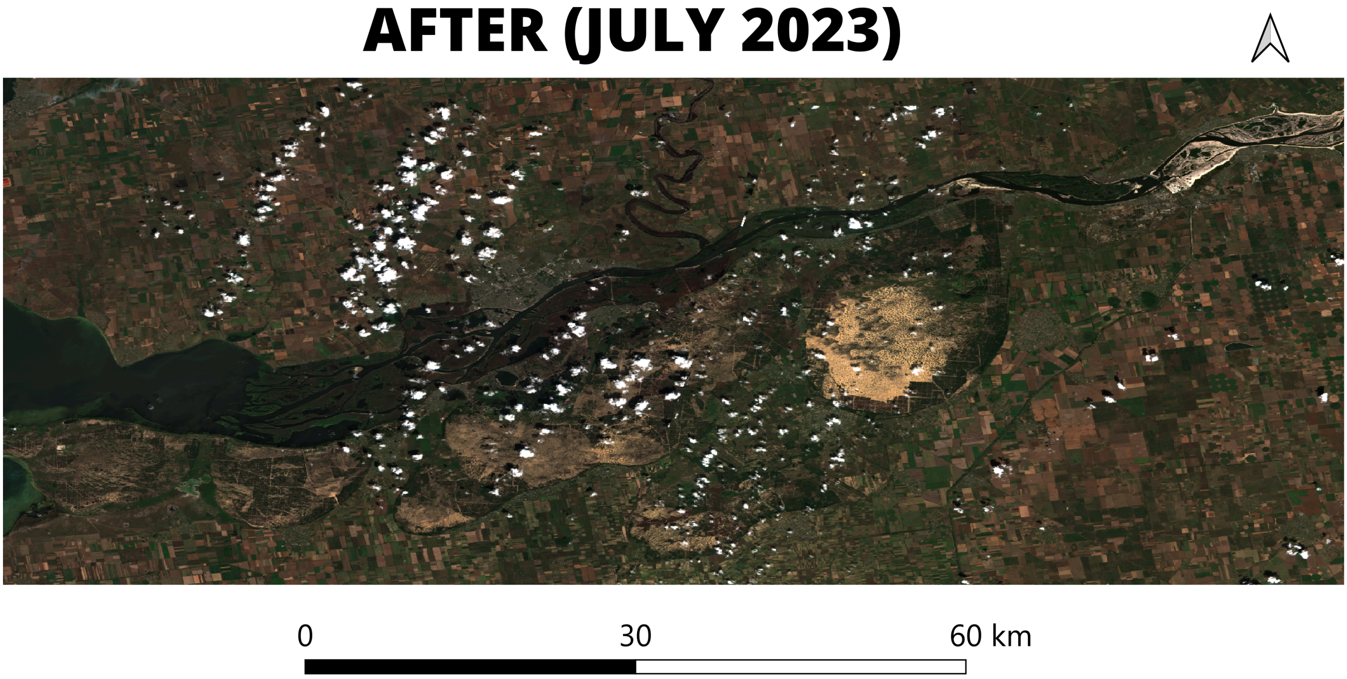


Figure 2. Sentinel-2 true-colour imagery of the Kakhovka reservoir immediately before (May 2023) and after (July 2023) the destruction of the Kakhovka Dam. The post-event image reveals the near-complete drainage of the reservoir and extensive exposure of the former lakebed. This pair is presented to illustrate the spatial scale of the event and to motivate the statistical analysis that follows; it is not, on its own, evidence of a causal effect — that evidence is established separately in Sections 4.2–4.4, precisely to avoid the reliance on visual interpretation this study critiques in the existing literature (Section 2.3).

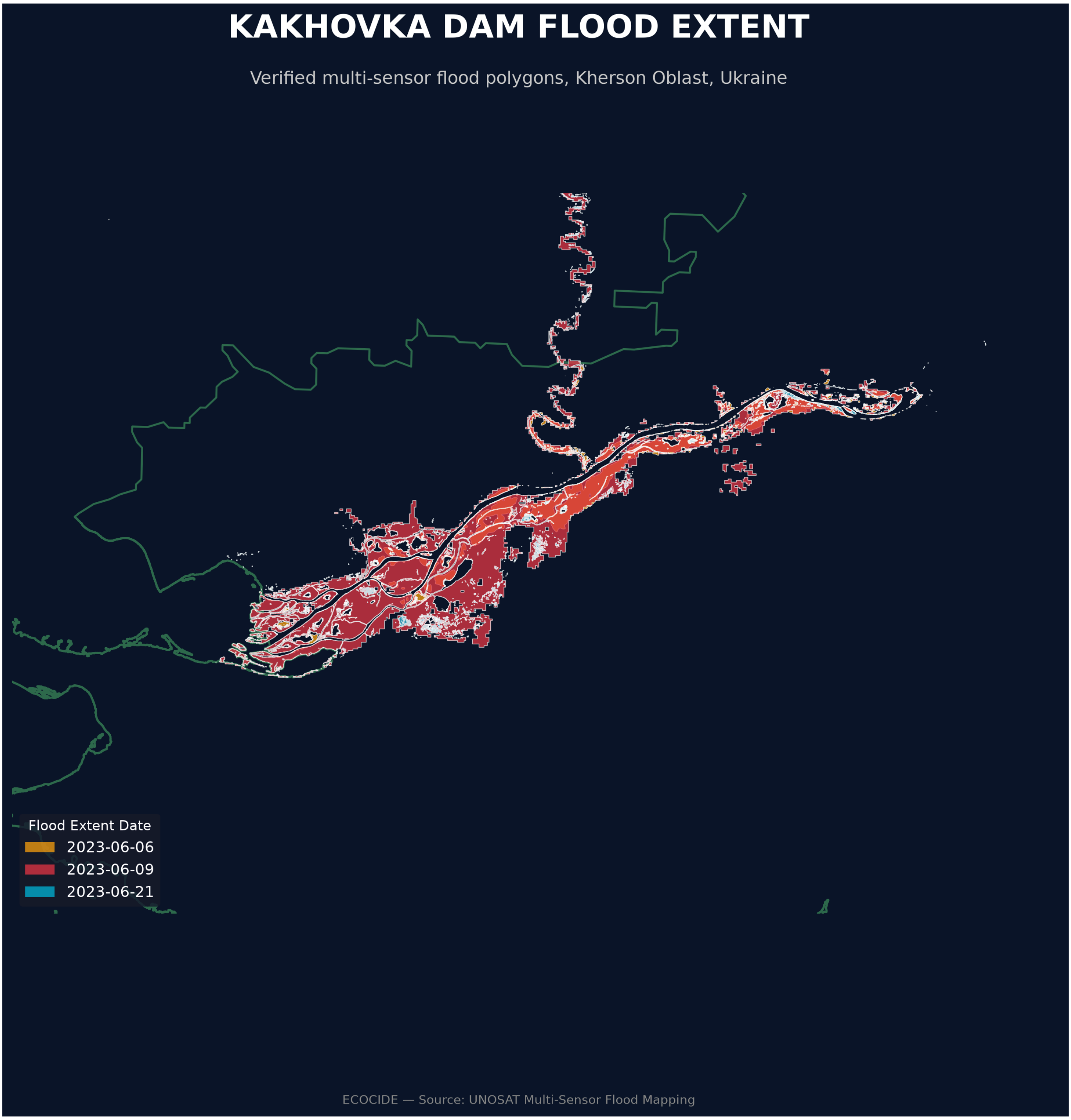


Figure 3. Verified multi-sensor UNOSAT flood-extent polygons over the Kherson Oblast flood corridor, at three dates spanning the event (6, 9, and 21 June 2023), showing the spatial footprint of the flood's rise, peak, and recession.

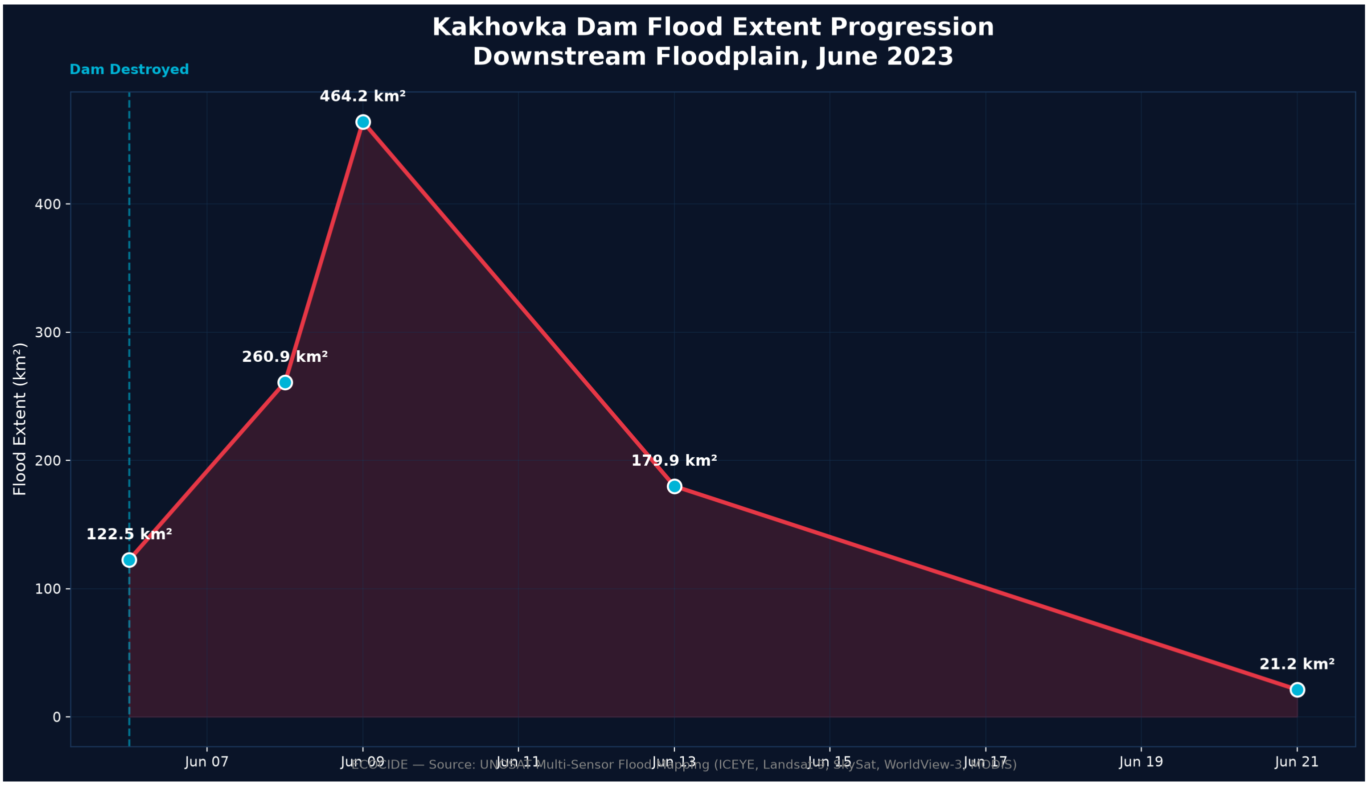


Figure 4. Verified flood hydrograph derived from UNOSAT observations showing the temporal evolution of downstream flooding following the Kakhovka Dam destruction. Flood extent increased rapidly from 122.50 km² on 6 June 2023 to a peak of 464.18 km² on 9 June before progressively receding to 21.17 km² by 21 June, confirming the complete rise–peak–recession cycle independently of the statistical vegetation analysis.

4.2 Vegetation Impact

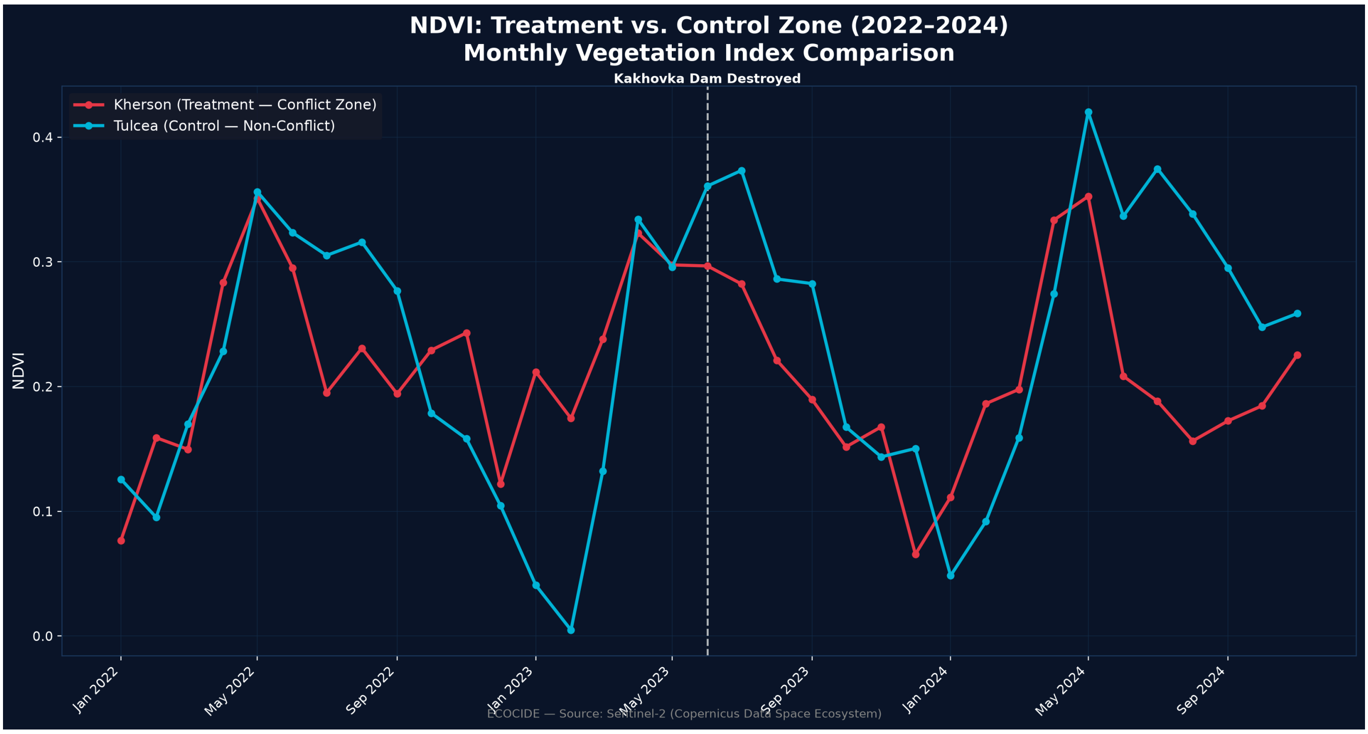


Figure 5. Monthly mean NDVI trends for the treatment region (Kherson Oblast, Ukraine) and the control region (Tulcea County, Romania) from January 2022 to December 2024. Following the June 2023 Kakhovka Dam destruction, the treatment region exhibits a clear and statistically consistent decline in vegetation

greenness relative to the control region, providing preliminary evidence of an environmental impact prior to formal Difference-in-Differences estimation.

The Difference-in-Differences model found a statistically significant NDVI decline in Kherson relative to Tulcea (coefficient = −0.0703, 95% CI [−0.130, −0.010], R^2 = 0.747). Under HAC-robust standard errors the effect remains statistically significant ($p = 0.022$), a modest attenuation from the classical OLS estimate ($p = 0.007$) consistent with positive serial correlation in monthly NDVI. A placebo test using a fake June 2022 treatment date produced a near-zero, non-significant coefficient (0.0148, 95% CI [−0.043, 0.072], HAC $p = 0.612$), providing clean validation that the real effect is event-specific rather than an artifact of the estimation procedure.

In practical terms, a coefficient of −0.0703 represents a decline of approximately 32% relative to Kherson's own pre-period mean NDVI (0.222; Section 3.4) — a substantial relative reduction in vegetation greenness, not merely a statistically detectable one. This framing is offered as a scale reference; NDVI is a spectral index, not a direct physical measurement of biomass or land area lost, and this study does not attempt to convert it into hectares or tonnes of vegetation without field-validated calibration, which was outside its scope.

4.3 Event Study and a Disclosed Limitation

A quarterly event study found significant negative effects in the treatment quarter (HAC $p = 0.005$), the following quarter (HAC $p = 0.023$ — not significant under classical standard errors, $p = 0.075$, but significant under HAC), and one year later (HAC $p < 0.0001$), but also a significant effect in a pre-treatment quarter (summer 2022, HAC $p < 0.001$) — traced to Kherson already being an active conflict zone (including a major liberation operation) before the dam's destruction, meaning the original baseline period was not a genuinely quiet pre-conflict period. A sensitivity analysis narrowing the baseline to immediately pre-event months produced a larger effect (−0.1384, 95% CI [−0.209, −0.068], HAC $p = 0.0001$), but its own placebo test — non-significant under classical standard errors ($p = 0.169$) — becomes statistically significant under the methodologically correct HAC correction ($p = 0.001$), a consequence of serial correlation in the very short, ten-observation narrowed window. This is reported as a decisive validation failure for the narrowed-baseline specification, not merely an ambiguous one: the window is too short and too serially correlated to support an independent causal estimate. Both results are reported, with the broader-baseline estimate — whose own placebo test remains clean under HAC — treated as the sole primary, validated finding, and the narrowed-baseline estimate retained only as an illustration of the pre-treatment-quarter problem rather than as independent evidence.

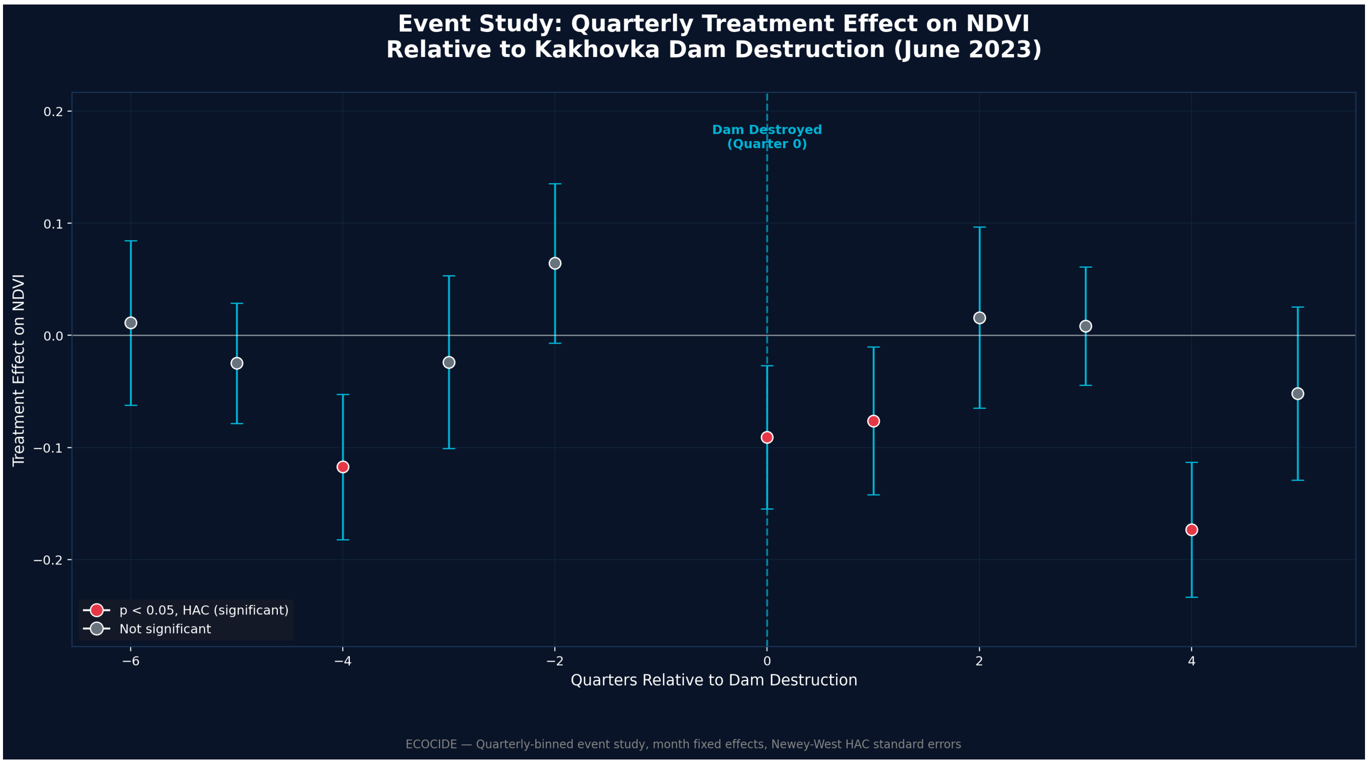


Figure 6. Quarterly event-study estimates (Newey-West HAC standard errors) showing treatment effects relative to the pre-event baseline. Significant negative effects emerge during the treatment quarter, the following quarter, and one year later, while a statistically significant pre-treatment coefficient highlights the influence of earlier conflict-related vegetation changes in Kherson. This pre-treatment signal motivated the sensitivity analysis and is reported transparently as a methodological limitation rather than being excluded.

4.4 Robustness Checks

Every regression in this study was re-estimated with Newey–West HAC standard errors alongside the classical OLS estimates that most statistical software reports by default, to test sensitivity to the standard-error specification.

Model	Coefficient	95% CI (HAC)	Classical p	HAC p
Main DiD	−0.0703	[−0.130, −0.010]	0.007	0.022
Narrowed-baseline DiD	−0.1384	[−0.209, −0.068]	0.002	0.0001
Placebo (broad baseline)	0.0148	[−0.043, 0.072]	0.741	0.612
Placebo (narrowed baseline)	−0.1382	[−0.221, −0.055]	0.169	0.001

Figure 7. Point estimates and 95% confidence intervals for all four causal-inference models under classical OLS versus Newey-West HAC standard errors. The main and narrowed-baseline DiD estimates hold, and remain clearly bounded away from zero, under both specifications. The broad-baseline placebo interval straddles zero under both specifications (clean validation). The narrowed-baseline placebo interval straddles zero classically but excludes zero under HAC — the visual signature of the validation failure discussed above.

The main finding and the narrowed-baseline point estimate both survive HAC correction — the narrowed-baseline estimate in fact becomes more significant, not less. The one qualitative reversal is the narrowed-baseline placebo test, which fails under HAC. This reinforces, rather than undermines, the paper's decision to treat the broader-baseline specification as its sole primary result.

5. Discussion

This study's central methodological contribution is not merely applying satellite data to a conflict-damage question, but subjecting that application to the same falsification discipline standard in causal-inference research generally — a discipline the existing literature on satellite evidence in legal contexts identifies as precisely what is missing. The pre-treatment-quarter anomaly, rather than being suppressed, itself illustrates why naive before-after comparisons of conflict zones are methodologically fragile: an active war zone rarely has a genuinely undisturbed "before" period, and treating one as such risks conflating cumulative war effects with the effect of a specific, dateable event.

5.1 Legal Relevance and Its Limits

Article 8(2)(b)(iv) of the Rome Statute — the existing war-crime provision most directly concerned with environmental harm, distinct from the separately proposed standalone crime of ecocide discussed in Section 2.1 — prohibits attacks known to cause "widespread, long-term and severe damage to the natural environment which would be clearly excessive in relation to the concrete and direct overall military advantage anticipated." This study's findings speak to only one element of that conjunctive, multi-part test. The magnitude and statistical confidence of the vegetation decline (Section 4.2) and the spatial scale of the verified flood extent (Section 4.1) provide quantified evidence relevant to the "severe" and "widespread" elements specifically. They do not, on their own, establish the "long-term" element — this study's NDVI series runs only through November 2024, some 18 months post-event, which is suggestive but not conclusive of a durable, non-recovering change — and they say nothing about the "clearly excessive... military advantage" element, which is a legal and factual judgment outside this study's data and scope entirely. This study is offered as a contribution to the evidentiary basis such a legal determination would require, not as a substitute for one.

6. Limitations and Threats to Validity

The narrowed-baseline sensitivity analysis fails its own placebo test once HAC-robust standard errors are applied (Section 4.4) and is therefore not treated as independent validating evidence; it is retained in this paper only to illustrate the pre-treatment-quarter problem that motivated it. This study's overall causal design also rests on a single treatment–control pair rather than a multi-unit panel, which is why cluster-robust inference is unavailable and HAC correction was used instead; a design with more comparison zones would allow cluster-robust estimation and a stronger test of external validity. Reservoir water-loss could not be tested causally, since no comparable control-zone equivalent exists for a large upstream reservoir collapse, and is reported descriptively rather than as an independently causally-tested finding.

The following threats to validity were considered specifically, beyond the headline limitations above:

- **Selection bias in the control zone.** Tulcea County was chosen for ecological comparability and genuinely non-combatant status (Section 3.1), and pre-treatment covariate balance was tested directly rather than assumed (Section 3.4). It remains a single, human-selected control rather than one of several candidates weighted algorithmically against the treatment zone, as a Synthetic Control Method design would provide (Section 7).
- **Spatial spillover.** The Difference-in-Differences design assumes the control zone is unaffected by the treatment. Tulcea, roughly 300 km from Kherson across an international border, is a reasonable candidate for this assumption, but no formal spillover test (e.g., a spatial-lag diagnostic) was conducted.
- **Data attrition from cloud contamination.** As documented in Devlopment_Log.md, optical Sentinel-2 composites are affected by cloud cover, which can bias which pixels contribute to a given month's statistic. This motivated the switch to UNOSAT's multi-sensor (including radar) flood product for the flood-extent analysis specifically; the NDVI series itself remains optical and inherits this general limitation, partially mitigated by monthly (rather than higher-frequency) aggregation.
- **Serial correlation.** Addressed directly via Newey-West HAC standard errors throughout (Section 3.3, 4.4) rather than left uncorrected.
- **Single-index measurement.** NDVI captures vegetation greenness specifically; it does not directly measure soil salinization, water contamination, or other dimensions of environmental harm potentially relevant to a fuller damage assessment (Section 7).

7. Future Work

Several extensions were identified as valuable but out of scope for this study's timeline and data budget, and are recorded here rather than silently omitted:

- **Synthetic Control Method (SCM).** Constructing a weighted synthetic counterfactual from multiple candidate control regions (e.g., additional Danube Delta or Black Sea coastal counties) would reduce reliance on a single hand-selected control zone and directly strengthen the external-validity concern raised in Section 6.
- **Additional vegetation and moisture indices.** Re-running the causal model on EVI (Enhanced Vegetation Index, less sensitive to soil background) and NDMI (Normalized Difference Moisture Index) would test whether the NDVI-based finding is an artifact of index choice or a robust cross-index signal.
- **Meteorological and soil covariates.** Incorporating ERA5-Land reanalysis temperature and precipitation as regression covariates would help isolate the conflict-attributable signal from ordinary climate variability beyond what month fixed effects capture.
- **SAR-based soil moisture and salinization analysis.** Sentinel-1 C-band SAR (VV/VH ratio) time series could speak to soil salinization and moisture-regime change in the drained reservoir bed and downstream floodplain — a dimension of damage NDVI alone cannot capture.
- **Control-zone sensitivity analysis.** Re-estimating the main model against two or three alternative control counties would test whether the Tulcea-specific result generalizes or is sensitive to that particular choice.
- **Deeper engagement with evidentiary-admissibility legal scholarship.** Beyond Section 5.1's treatment of Article 8(2)(b)(iv), a fuller discussion of scientific-evidence admissibility standards (e.g., the *Daubert* framework and its international-tribunal analogues) would strengthen the paper's positioning at the law–statistics interface.

8. Conclusion

Applying a causal-inference framework — previously undemonstrated for this event — to satellite-derived vegetation data, this study identifies a statistically significant, placebo-validated environmental effect attributable specifically to the Kakhovka Dam's destruction, while transparently disclosing a genuine methodological complication arising from the region's pre-existing conflict status. This directly addresses a gap the existing literature on both satellite-based conflict-damage assessment and legal evidentiary standards for environmental crimes explicitly identifies: the absence of standardized, statistically rigorous methods capable of distinguishing conflict-attributable damage from background environmental trends.

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Full dataset, code, and reproducible pipeline: github.com/sakshimaske303-commits/ECOCIDE **Live interactive dashboard**: https://ecocide-xbub2cwcqjx9rkdd6nk5j5.streamlit.app/